Semantic Entropy Formalism

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1 Semantic entropy

We treat semantic entropy $E(\sigma)$ as a nonnegative scalar that captures representational underspecification, ambiguity, and contradiction within a sphere. It admits the following properties:

1. Nonnegativity: $E(\sigma) \geq 0$.
2. Monotonicity under naive merge: without constraints, naive merges may increase entropy.
3. Controlled growth: permitted operations declare an entropy budget limiting increase.

2 Entropy accounting

Each transformation r advertises $\epsilon_r \geq 0$. For mediated merges we require the existence of reconciliation chains R_a, R_b such that the merged sphere respects the merge budget.

3 Metrics

Define modality divergence $\delta(M_1(k), M_2(k))$ for each modality (e.g., normalized edit distance for text, embedding distance for vectors). The full metric:

$$d(\sigma_1, \sigma_2) = |E(\sigma_1) - E(\sigma_2)| + \sum_{k \in \mathcal{K}} \delta(M_1(k), M_2(k)).$$